

ARTI406 - Assignment 2.ipynb

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..sample_dataairbnb_top_cities.csv

[1]✓ 1s

Import Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")

1. Load Dataset

[2]✓ 2s

pd.set_option("display.max_columns", None)

df = pd.read_csv("airbnb_top_cities.csv")
df.head()

/tmp/ipykernel_2622/3989594923.py:3: DtypeWarning: Columns (4,17) have mixed types. Specify dtype option on import or set low_memory=False.
df = pd.read_csv("airbnb_top_cities.csv")

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type	price	minimum_nights	number_of_reviews	last_
0	13913	Holiday London DB Room Let-on going	54730	Alina	NaN	Islington	51.56861	-0.11270	Private room	70.0	1	55	202
1	15400	Bright Chelsea Apartment, Chelsea	60302	Philippa	NaN	Kensington and Chelsea	51.48780	-0.16813	Entire home/apt	149.0	4	97	202

Variables Terminal

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2. Data Quality Assessment

2.1 Check Data Types

Data types must match the real meaning of each column. For example:

- Date should be datetime
- Revenue should be numeric

[3] ✓ 0s

df.dtypes

id	int64
name	object
host_id	int64
host_name	object
neighbourhood_group	object
neighbourhood	object
latitude	float64
longitude	float64
room_type	object
price	float64

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scrape_dateobject

dtype: object

We observe that some columns are stored as object type. Date and Amount may not yet be in the correct format. Incorrect data types can prevent proper calculations and time-based analysis.

2.2 Convert Incorrect Data Types

We will convert:

- Date to datetime format
- Amount to numeric after removing currency symbols

[*] 0sdf['last_review'] = pd.to_datetime(df['last_review'])df['scrape_date'] = pd.to_datetime(df['scrape_date'])df.dtypes

id	int64
name	object
host_id	int64
host_name	object

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id	int64
name	object
host_id	int64
host_name	object
neighbourhood_group	object
neighbourhood	object
latitude	float64
longitude	float64
room_type	object
price	float64
minimum_nights	int64
number_of_reviews	int64
last_review	datetime64[ns]
reviews_per_month	float64
calculated_host_listings_count	int64
availability_365	int64
number_of_reviews_ltm	int64
license	object
city	object

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numerical calculations correctly.

3. Handling Missing Values

3.1 Detect Missing Values

Missing values reduce data quality and can affect model performance.

```
[5]: df.isna().sum()
```

	0
id	0
name	0
host_id	0
host_name	90
neighbourhood_group	273392
neighbourhood	0
latitude	0
longitude	0
room_type	0
price	152852
minimum_nights	0



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longitude	0
room_type	0
price	152852
minimum_nights	0
number_of_reviews	0
last_review	66365
reviews_per_month	66365
calculated_host_listings_count	0
availability_365	0
number_of_reviews_ltm	0
license	152949
city	0
scrape_date	0

dtype: int64

The output shows whether any column contains missing values. If all values are zero, the dataset is complete. If any column contains missing values, we must handle them.

3.2 Demonstration: Introduce Artificial Missing Values

Why?

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```
df_missing.loc[0:5, 'Amount'] = np.nan
```

- `df_missing` : The pandas DataFrame you are modifying.
- `.loc[0:5, 'Amount']` : This uses the label-based indexer to select specific rows and columns.
- `0:5` : Selects rows with index labels 0, 1, 2, 3, 4, and 5. In label-based indexing, the end index is inclusive.
- `'Amount'` : Selects the column named 'Amount'.
- `= np.nan` : Assigns the value `np.nan` (which stands for "Not a Number") to all the selected cells. This is the standard way to represent missing or null values in numerical columns in pandas. The column's data type will be converted to float

[6]
✓ On

```
df_missing = df.copy()

df_missing['price'] = df_missing['price'].fillna(df_missing['price'].median())
df_missing['reviews_per_month'] = df_missing['reviews_per_month'].fillna(0)

df_missing['host_name'] = df_missing['host_name'].fillna('Unknown')
df_missing['neighbourhood_group'] = df_missing['neighbourhood_group'].fillna('Unknown')
df_missing['license'] = df_missing['license'].fillna('Unknown')

df_missing = df_missing.dropna(subset=['last_review'])

df_missing.isna().sum()
```

```
0
id      0
name    0
host_id 0
```

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id	0
name	0
host_id	0
host_name	0
neighbourhood_group	0
neighbourhood	0
latitude	0
longitude	0
room_type	0
price	0
minimum_nights	0
number_of_reviews	0
last_review	0
reviews_per_month	0
calculated_host_listings_count	0
availability_365	0
number_of_reviews_ltm	0
license	0

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dtype: int64

Now the Amount column contains missing values.

```
[7]: print("Original shape:", df.shape)
      print("After handling missing values:", df_missing.shape)
```

Original shape: (292802, 20)
After handling missing values: (226437, 20)

```
[8]: df_missing.head(10)
```

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type	price	minimum_nights	number_of_reviews
0	13913	Holiday London DB Room Let-on going	54730	Alina	Unknown	Islington	51.56861	-0.11270	Private room	70.0	1	55
1	15400	Bright Chelsea Apartment, Chelsea	60302	Philippa	Unknown	Kensington and Chelsea	51.48780	-0.16813	Entire home/apt	149.0	4	97
2	17402	Very Central Modern 3-Bed/2 Bath By Oxford St W1	67564	Liz	Unknown	Westminster	51.52195	-0.14094	Entire home/apt	411.0	3	56
3	24328	Battersea live/work artist house	41759	Joe	Unknown	Wandsworth	51.47072	-0.16266	Entire home/apt	163.0	7	95
4	36274	Bright 1 bedroom apt off brick lane in	133271	Hendryks	Unknown	Tower Hamlets	51.52322	-0.06979	Entire home/apt	210.0	5	16

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1	15400	Bright Chelsea Apartment, Chelsea	60302	Philippa	Unknown	Kensington and Chelsea	51.48780	-0.16813	Entire home/apt	149.0	4	97
2	17402	Very Central Modern 3-Bed/2 Bath By Oxford St W1	67564	Liz	Unknown	Westminster	51.52195	-0.14094	Entire home/apt	411.0	3	56
3	24328	Battersea live/work artist house	41759	Joe	Unknown	Wandsworth	51.47072	-0.16266	Entire home/apt	163.0	7	95
4	36274	Bright 1 bedroom apt off brick lane in Shoreditch	133271	Hendryks	Unknown	Tower Hamlets	51.52322	-0.06979	Entire home/apt	210.0	5	15
5	36299	Kew Gardens 3BR house in cul-de-sac	155938	Geert	Unknown	Richmond upon Thames	51.48145	-0.28107	Entire home/apt	280.0	3	116
6	36660	You are GUARANTEED to love this	157884	Agri & Roger	Unknown	Haringey	51.58478	-0.16057	Private room	90.0	2	730
7	38605	SUNNY ROOM PRIVATE BATHROOM PLUS BREAKFAST	165579	Elisa & Dom	Unknown	Hammersmith and Fulham	51.50681	-0.23345	Private room	61.0	2	387
8	38610	Short Term Home	165579	Elisa & Dom	Unknown	Hammersmith and Fulham	51.50701	-0.23362	Entire home/apt	340.0	91	42
9	38995	SPACIOUS ROOM IN CONTEMPORARY STYLE FLAT	167281	Cesar	Unknown	Southwark	51.47860	-0.06114	Private room	49.0	1	72